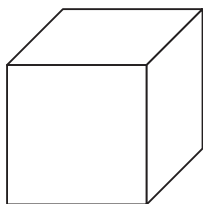


PRACTICE TEST

For questions 1 and 2, choose the correct answer: A, B, C, or D

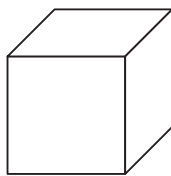
1. The volume V cubic inches of each cube is given. For which cube is the edge length an irrational number?

A.



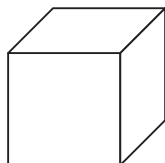
$$V = 125 \text{ in.}^3$$

B.



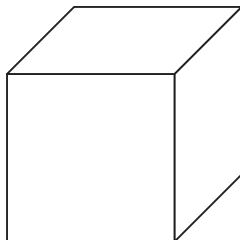
$$V = 75 \text{ in.}^3$$

C.



$$V = 64 \text{ in.}^3$$

D.



$$V = 216 \text{ in.}^3$$

2. Which number is rational?

A. $\sqrt{0.09}$

B. $\sqrt{50}$

C. $\sqrt[3]{-\frac{64}{121}}$

D. π

3. a) Which is greater, $\sqrt{70}$ or $5\sqrt{3}$? Justify your answer.
b) Sketch a number line to illustrate the numbers in part a.

4. Evaluate without using a calculator.

a) $\sqrt[4]{\frac{256}{81}}$

b) $(-4)^{-2}$

c) $0.81^{\frac{3}{2}}$

d) $16^{-\frac{1}{2}}$

5. Write $44^{\frac{1}{2}}$ as a radical in simplest form.

6. A student simplified $\frac{x^{-1}y^3}{xy^{-2}}$ as follows:

$$\begin{aligned}\frac{x^{-1}y^3}{xy^{-2}} &= x^{-1+1} \cdot y^{3-2} \\ &= x^0y^1 \\ &= y\end{aligned}$$

Is the student correct? If not, describe any errors and write a correct solution.

7. Simplify each expression. Write your answers using positive exponents.

a) $(p^{-2}q^{-1})^2(pq^{\frac{1}{2}})^2$

b) $\left(\frac{c^6d^5}{c^3d^4}\right)^{-\frac{1}{3}}$

8. Scientists use the formula $d = 0.099m^{\frac{9}{10}}$ to calculate the volume of water, d litres, that a mammal with mass m kilograms should drink in 1 day. Calculate how much water a 550-kg moose should drink in one day.